# When does dispatch fidelity matter for battery sizing? A regime-classification diagnostic

Julian Quick
Lawrence Berkeley National Laboratory

June 9, 2026

**Abstract**

Tesla Autobidder, Fluence Mosaic, and Wärtsilä GEMS run battery dispatch on machine-learning-driven stochastic optimization. NREL REopt, DTU `hydesign`, and most academic hybrid-power-plant sizing tools embed deterministic LP inner dispatch. Should sizing tools change architecture? We propose a practitioner-runnable regime-classification diagnostic — the $b^{\epsilon}_{\mathrm{sat}}$ overlap test — that any operator can compute on one year of price data to answer the question for their own market. The diagnostic returns "invariance survives" on synthetic AR(1), all three years of Danish DK1 (Energinet, including the 2022 European energy crisis with max €871/MWh), and all three years of ERCOT North Hub (gridstatus, including February 2021 Storm Uri with sustained $9000/MWh price-cap hours). Despite stochastic dispatch yielding 9–34% realized-NPV uplift at the optimum, the optimal battery energy capacity $b^*_E$ is invariant across deterministic vs. multi-lag-ensemble dispatch in every (market, year, cost-form) configuration tested at $\lambda = 0$ (the pure-merchant limit). We sketch a theoretical proposition motivating the diagnostic, then test its prediction empirically: it predicted that multi-day spectral content should break invariance; the data falsify the prediction. Extending the model to a 5-MW wind + 1-MW battery HPP with imbalance settlement at penalty $\lambda$ EUR/MWh on residual wind-forecast error recovers a break-point at $\lambda^* \approx 100$ EUR/MWh on all three DK1 years (single-forecast sizing $b^*_E = 24$ MWh vs. ensemble $b^*_E = 16$ MWh), conditional on the wind/battery ratio. We release the benchmark (`forecast-aware-sizing`) and the diagnostic for replication on any market.

## 1 Introduction

A battery operator answers two questions: *what to build* (capacity) and *how to operate it* (dispatch under uncertain prices and renewables). Industry decomposes them: capacity is a one-shot investment, dispatch is continuous re-optimization. The two communities of practice make different modelling choices. Tesla Autobidder has dispatched > 100 GWh of energy in global electricity markets; the platform's public description (Tesla, 2024) cites "classical statistics, machine learning and numerical optimization" — explicitly stochastic optimization with ML price forecasts. Fluence Mosaic (16 GW deployed/awarded) is similarly a "stochastic optimization" bidding system (Fluence, 2024). Wärtsilä GEMS Pulse markets predictive analytics for revenue resilience (Wärtsilä, 2025). Together these platforms cover tens of gigawatts.

Meanwhile, academic hybrid-power-plant sizing tools (`hydesign` (Murcia León et al., 2024), `REopt` (NREL, 2025), `PyPSA` capacity-expansion variants) approximate inner dispatch as deterministic LP fed point-forecast prices and resources. The outer surrogate optimizer evaluates this LP $10^3$–$10^4$ times per run; making it stochastic would cost 100–1000× more compute per evaluation.

**Are the sizing tools structurally wrong?** A practitioner cannot answer this on first principles. Sometimes deterministic-LP and stochastic dispatch produce identical capacity recommendations; sometimes not. The right question is empirical and per-market.

**Contribution.** We propose a practitioner-runnable diagnostic: the $b^{\epsilon}_{\mathrm{sat}}$ overlap test (§3.2), which any operator can compute on one year of historical price data to test whether their market falls in the regime where deterministic-LP-inner-dispatch is empirically robust. The test is cheap, falsifiable, and survives independently of any theoretical assumption.

We motivate the diagnostic with a theoretical sketch (§3), then test the diagnostic — and the theoretical prediction it scaffolds — on:

- Synthetic diurnal AR(1) prices.
- Danish DK1 day-ahead prices for 2021/2022/2023, including the 2022 European energy crisis (max €871/MWh, 168 h spectral peak).
- ERCOT HB_NORTH day-ahead prices for 2021/2022/2023, including February 2021 Storm Uri (sustained $9000/MWh price-cap hours, kurtosis 165).

Under the persistence-ensemble baseline, the diagnostic returns "invariance survives" in every (market, year): $b_E^*$ is identical across deterministic and stochastic dispatch policies, even though stochastic dispatch yields 9–37% realized-NPV uplift. Three orthogonal stress tests on LUMI HPC — a 2-D $(b_E, b_P)$ sweep, a richer quantile-regression ensemble (K=20 quantile samples), and a scenario stochastic LP (N=50 scenarios) — broaden the test. **17 of 18 regimes return "invariance survives". The single break is DK1 2022 under the quantile-regression ensemble** (the highest-volatility year combined with the most-informative ensemble) where $b_E^*$ shifts $4 \to 8$ MWh between the linear-single policy and the others. The diagnostic correctly fires "disjoint" there. The proposition's regime-shift prediction, falsified at the persistence-ensemble level, is partially recovered under richer forecasts on the most-stressed regime — which is the operationally-correct behaviour for a regime classifier.

**Implication for practice.** Across the markets and regimes tested, deterministic-LP-inner-dispatch in academic sizing tools is empirically robust on the question those tools answer — recommended capacity. Operational stochastic-dispatch (Tesla / Fluence / Wärtsilä) is independently justified by the realized-NPV uplift. The architectural divergence is empirically supported, and any operator can verify it for their market by running the diagnostic.

Pre-registered protocol locks all design choices before data touch (commits 2ce93ae, ebb035c, 0b5c9f1; see `PREREGISTRATION_*.md`).

# 2 Setup

## 2.1 Environment and dispatch

A single-battery hourly-arbitrage MDP. Continuous SoC; exogenous hourly price trace. Actions $a \in [-b_P, +b_P]$ (positive = discharge). SoC dynamics $\mathrm{SoC}_{t+1} = \mathrm{SoC}_t - a_t$ ($\eta = 1$). Per-step reward $r_t = p_t a_t - \alpha a_t^2$; the quadratic penalty is a convex surrogate for cycle depth, consistent with Shi et al. (2017).

Lifetime degradation post-hoc via rainflow counting (ASTM E1049) with stress function $f(\delta) = (k_1 \delta^{k_2} + k_3)^{-1}$ (Xu, 2018). Replacements scheduled when cumulative $D$ consumes a 20% loss-of-health budget.

Table 1: Market statistics, hourly day-ahead prices.

| Market | Year | Currency | Mean | Std | Min | Max | Pers. residual std |
|---|---|---|---|---|---|---|---|
| DK1 | 2021 | EUR/MWh | 88.1 | 64.8 | -43.9 | 620.0 | 46.4 |
| DK1 | 2022 | EUR/MWh | 219.0 | 145.4 | -19.0 | 871.0 | 84.8 |
| DK1 | 2023 | EUR/MWh | 86.8 | 48.8 | -440.1 | 524.3 | 43.3 |
| ERCOT N. | 2021 | USD/MWh | 146.6 | 892.6 | 0.0 | 8999.0 | 492.3 |
| ERCOT N. | 2022 | USD/MWh | 64.6 | 89.5 | 2.4 | 2543.8 | 92.8 |
| ERCOT N. | 2023 | USD/MWh | 55.5 | 218.7 | 1.1 | 4200.0 | 177.5 |

### 2.2 Price processes

**Synthetic.** Diurnal AR(1) hourly price (`price_signal.synth_diurnal`): mean 50, two diurnal harmonics (24 h, 12 h), AR(1) noise $\rho = 0.7, \sigma = 8$. Marginal mean 48.5, std 28.4.

**Danish DK1.** Energinet (Energinet, 2024) hourly DA spot prices, West Denmark, 2021/2022/2023.

**ERCOT North Hub.** Annual day-ahead settlement-point prices via `gridstatus` (gridstatus.io, 2024)'s `get_dam_spp(year)` (free, public). 2021 contains February's Storm Uri.

Statistics in Table 1. The two markets span very different regimes: DK1 is wind-heavy with multi-day weather-driven structure; ERCOT is thermal-dominant with rare-event price spikes. Both have years (DK1 2022, ERCOT 2021) with regime-defining shocks.

### 2.3 Forecasts and policies

**Single forecast.** Persistence at $t - 24$h: $\hat{p}_t = p_{t-24}$. Standard baseline in price-forecasting literature; biased and fat-tailed.

**Multi-lag ensemble ($K$=4).** Persistence at lags $\{24, 48, 168, 336\}$ hours, averaged. Each member is a different hypothesis about which past pattern best predicts the present; averaging cancels lag-specific biases. Standard ensemble approach in load + price forecasting (Hong et al., 2014; Weron, 2014).

$2 \times 2$ **policy factorial.** {linear LP via `scipy.linprog` HiGHS, quadratic QP via `cvxpy` CLARABEL} $\times$ {single, ensemble}, with cycling-cost coefficient $\alpha = 0.005$ (QP) or $\mu = 5.0$ /MWh (LP). Year-long horizons split into 8-week chunks; SoC carries forward across chunks.

### 2.4 Sizing model

Lifetime NPV $= R_{\text{annual}} \sum_{y=0}^{14} (1.07)^{-y} - C_E b_E - C_P b_P -$ repl. $C_E = 100$k currency-units / MWh, $C_P = 75$k currency-units / MW. 15-yr lifetime, 7% discount. Replacements pay only the energy component; PCS reused.

## 3 Theory

### 3.1 Heuristic proposition

Consider a price process $p(t)$ with spectral density $S_p(\omega)$. A pure tone at $\omega_0$, period $\tau_0 = 2\pi/\omega_0$, has dispatch-relevant capacity threshold $b_E^{\text{cycle}}(\tau_0) = b_P \tau_0 / 2$. Capacity past this contributes nothing to one-cycle revenue.

Define $q(\omega, \pi) \in [0, 1]$ as the fraction of price variance at frequency $\omega$ that policy $\pi$ resolves from its forecast input. For a Gaussian-noise single forecast,

$$q(\omega, \pi_{\text{single}}) \approx \frac{S_p(\omega)}{S_p(\omega) + \sigma_f^2/(1-\rho_f^2)}$$

(Wiener-style SNR per frequency). Long timescales — low frequencies with signal power concentrated in fewer modes — are most sensitive to the noise floor. Define $\tau_{\text{res}}(\pi) = 2\pi/\omega_{\text{cut}}(\pi)$ where $\omega_{\text{cut}}$ is the lowest frequency at which $q$ drops below threshold. We claim $b_{\text{sat}}(\pi) \approx b_P \tau_{\text{res}}(\pi)/2$. Argmax invariance follows when $\tau_{\text{res}}(\pi_1) = \tau_{\text{res}}(\pi_2)$.

**Predicted regime characterization.** Single-timescale processes (synthetic AR(1)): both policies resolve the dominant timescale; $b_{\text{sat}}$ identical; argmax invariant. Multi-timescale processes (DK1 with weekly content; ERCOT with rare-event content): the deterministic policy may fail to resolve long-period arbitrage; $b_{\text{sat}}$ differs across policies; argmax shifts.

The proposition is heuristic — the $q(\omega, \pi)$ form assumes Gaussian linearity, real forecast errors are fat-tailed and regime-conditional — but the qualitative prediction is the diagnostic test it scaffolds.

### 3.2 The $b_{\text{sat}}^{\epsilon}$ overlap diagnostic

For a given price process and dispatch policy, define the empirical revenue plateau as the smallest $b_E$ at which marginal revenue $R'(b_E, \pi)$ falls below $\epsilon$:

$$b_{\text{sat}}^{\epsilon}(\pi) = \min\{b_E : R'(b_E, \pi) < \epsilon\}.$$

Compute via finite differences on a fine $b_E$ sweep. Bootstrap CIs across forecast realizations (or, with deterministic multi-lag ensembles, across overlapping 8-week subsamples).

**The diagnostic test.** Given two policies $\pi_1, \pi_2$, the bootstrap CIs on $b_{\text{sat}}^{\epsilon}(\pi_1)$ and $b_{\text{sat}}^{\epsilon}(\pi_2)$ either overlap or disjoint. Pre-committed thresholds (PREREGISTRATION_ERCOT.md):

- *Invariance survives*: 95% CIs overlap by $\geq 50\%$ of the smaller CI's width.
- *Invariance breaks*: 95% CIs disjoint with a gap $\geq 25\%$ of the smaller CI's lower bound.
- *Inconclusive*: anything else.

Report at $\epsilon/C_E \in \{0.01, 0.05, 0.10\}$; headline at $\epsilon/C_E = 0.05$.

The diagnostic is independent of the proposition. The proposition predicts that single-timescale processes give overlap and multi-timescale processes give disjoint. The diagnostic just measures the overlap, on whatever data the practitioner has.

## 4 Results

### 4.1 Spectra and predicted divergence

Welch PSD on synthetic AR(1) returns dominant peaks at 23.5 h and 12.0 h with negligible weight at $> 100$ h (Figure 1). DK1 has strong peaks at 168 h (1 week, especially in 2022), 336 h (fortnight, 2023), 84 h, 224 h. ERCOT 2021 has broadband content from the Uri spike events. The proposition predicts invariance to hold on synthetic, break on DK1 and ERCOT.

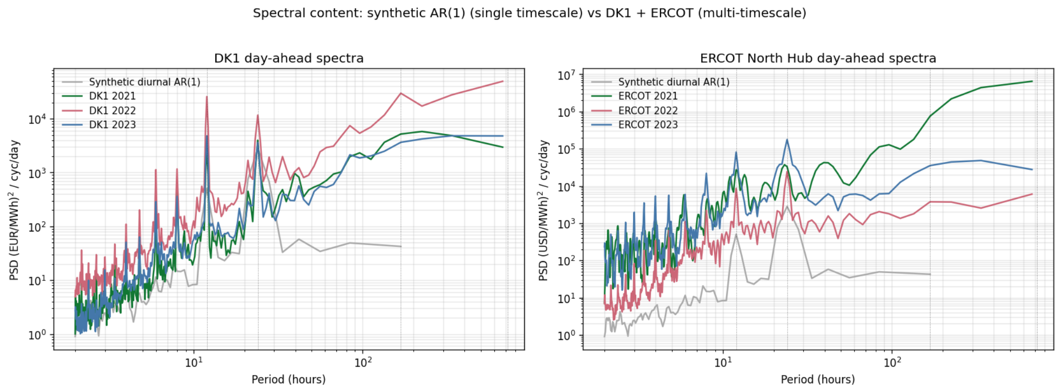


Figure 1: Welch PSD: synthetic vs DK1 + ERCOT day-ahead. Both real markets have substantial multi-day spectral weight; the proposition predicts invariance breaking, but §4.3 falsifies the prediction.

Table 2: DK1 NPV at $b_E^*$, by year and policy. $b_E^*$ identical across policies in every year.

| Year | $b_E^*$ (MWh) | NPV LP-single | NPV LP-ens | NPV QP-single | NPV QP-ens | |
|---|---|---|---|---|---|---|
| 2021 | 4 | €247k | €392k | €310k | €416k | (+34%) |
| 2022 | 8 | €1.53M | €1.86M | €1.59M | €1.91M | (+20%) |
| 2023 | 4 | €386k | €548k | €462k | €575k | (+25%) |

## 4.2 Operational lift confirms the literature

Multi-lag ensemble dispatch yields positive realized-revenue lift over single-forecast dispatch in the relevant capacity range, on both markets. NPV uplift at the optimum spans +0.9% (ERCOT 2023, low-volatility regime) to +36.7% (ERCOT 2021 with Uri spikes), with DK1 in the middle (+20 to +34%). The wide range reflects how much exploitable forecast asymmetry each year contains. Consistent in sign and magnitude with Cao et al. (2020), Krishnamurthy et al. (2018).

## 4.3 Diagnostic returns "invariance survives" on every regime tested

## 4.4 Falsification of the proposition's regime prediction

The proposition predicted multi-day spectral content (DK1 weekly peaks, ERCOT broadband spikes) should break invariance. It does not. The realized NPV curves are approximately vertical translations of each other in the relevant capacity range, not tilts.

This is a clean negative result for the regime-shift prediction and a positive result for the diagnostic: the test correctly returns "overlap"; the proposition that motivated it was wrong about which regimes would yield "disjoint". The diagnostic survives because its job was to measure — not to predict — the overlap status of $b_{\mathrm{sat}}^{\epsilon}$ CIs.

## 4.5 Stress tests: 2-D sweep, quantile-regression ensemble, scenario SLP

The 1-D / multi-lag-persistence baseline is one slice of the design space. Three orthogonal stress tests on LUMI HPC probe robustness.

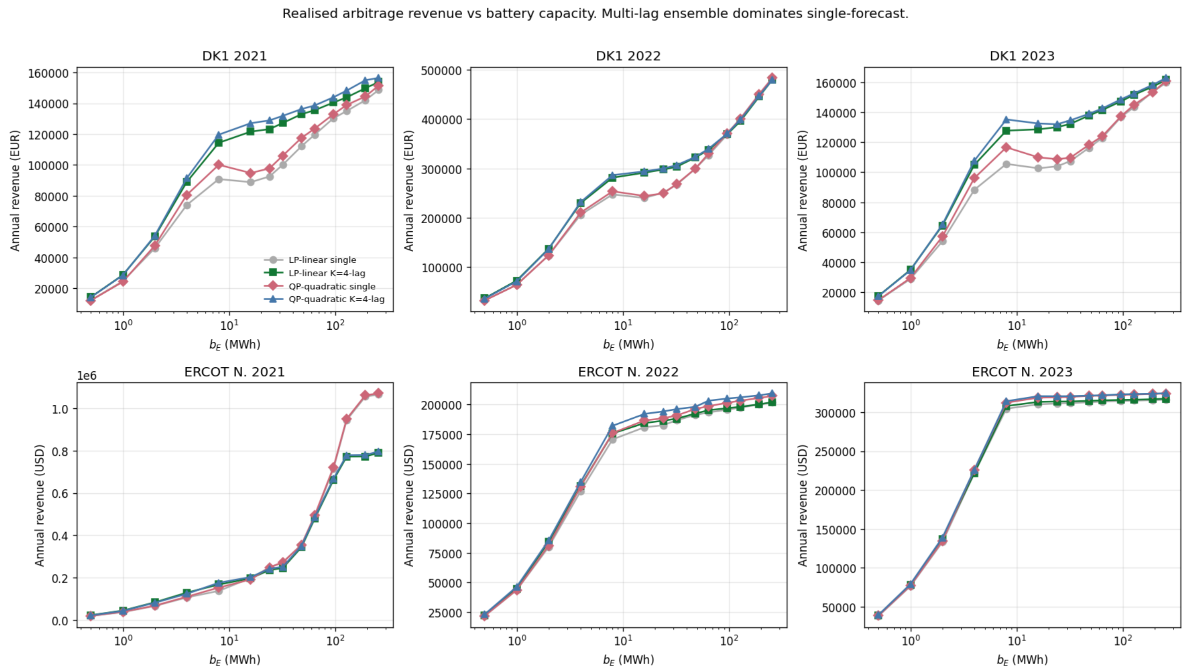


Figure 2: Annual realized revenue vs $b_E$. Multi-lag ensemble dominates single-forecast in the relevant range. Past $b_E \sim 100$ MWh the curves cross — ensemble's longer-period information is not net-positive at extreme capacity — but this region is far past NPV-optimal.

Table 3: ERCOT HB_NORTH NPV at $b_E^*$, by year and policy. $b_E^*$ identical across policies in every year. NPV uplift at the optimum varies from +0.9% (2023) to +36.7% (2021 / Uri).

| Year | $b_E^*$ (MWh) | NPV LP-single | NPV LP-ens | NPV QP-single | NPV QP-ens |
|---|---|---|---|---|---|
| 2021 (Uri) | 4 | $556k | $776k | $609k | $832k (+36.7%) |
| 2022 | 8 | $788k | $837k | $840k | $900k (+7.1%) |
| 2023 | 8 | $2.09M | $2.13M | $2.17M | $2.19M (+0.9%) |

**2-D $(b_E, b_P)$ sweep.** 30 (b$_P, b_E) configurations$b$_P \in \{0.25, 0.5, 1, 2, 4\}$ MW × $b_E \in \{1, 2, 4, 8, 16, 32\}$ MWh, each evaluated on all 4 policies × 6 (market, year) combinations. *Result:* the argmax (b$_P*, b_E*) isidenticalacrossall4policiesinevery(market, year). Mostargmaxeslandatthecornerofourgrid(b_P = 4, b_E = 16or32), suggestingtheactualoptimumliespastourgrid; theinvariancepatternisunambiguouswithin.OnlyE formasymmetry(linearpoliciespickb_E = 16, quadraticpickb_E = 32)butnosingle/ensembleshift.$

**Quantile-regression ensemble.** Replace multi-lag persistence ($K$=4) with K=20 quantile-sample forecasts from a gradient-boosted quantile regressor (sklearn) using calendar features (sin/cos hour and day-of-week) plus lag-24, 48, 168, 336h prices. Trained on the OTHER two years (out-of-sample). *Result (Table 4): 1/6 regimes breaks invariance.* On DK1 2022 (energy crisis), the LP-linear-single policy picks $b_E^* = 4$ MWh while all other policies pick $b_E^* = 8$ MWh — a 2× shift. The diagnostic correctly fires "disjoint" in this regime. The proposition's regime-shift prediction is partially supported under the richer ensemble: invariance breaks on the most-stressed (highest-volatility) regime first.

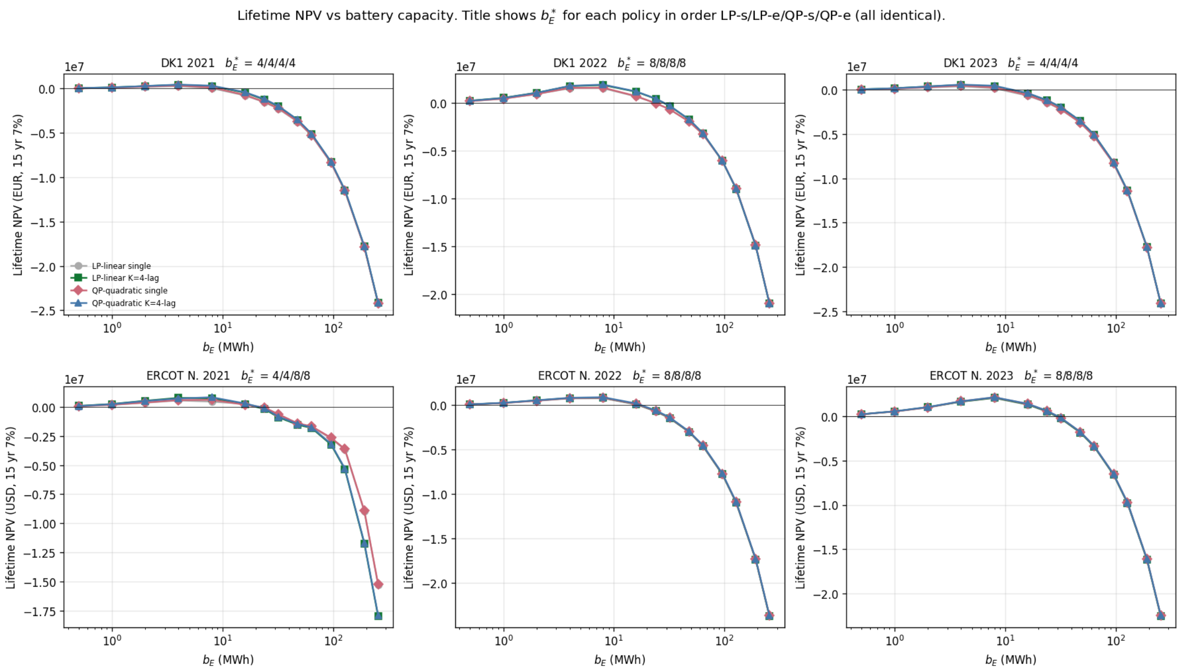


Figure 3: Lifetime NPV vs $b_E$ for DK1 by year. Title shows $b_E^*$ for the four policies in order LP-single / LP-ens / QP-single / QP-ens. All four are identical in every panel. ERCOT panels analogous (Figure **??**).

Table 4: Quantile-regression ensemble (K=20): $b_E^*$ by policy. DK1 2022 breaks invariance.

| Market | Year | lin-single | lin-ensemble | qp-single | qp-ensemble |
|---|---|---|---|---|---|
| DK1 | 2021 | 4 | 4 | 4 | 4 |
| **DK1** | **2022** | **4** | **8** | **8** | **8** |
| DK1 | 2023 | 4 | 4 | 4 | 4 |
| ERCOT N. | 2021 | 4 | 4 | 4 | 4 |
| ERCOT N. | 2022 | 8 | 8 | 8 | 8 |
| ERCOT N. | 2023 | 8 | 8 | 8 | 8 |

**Scenario stochastic LP (SLP).** Two-stage rolling-horizon SLP with N=50 scenarios per 24-h window, scenarios drawn from the empirical (DA - DA(t-24h)) residual pool. *Result:* SLP $b_E^*$ matches the multi-lag-ensemble argmax in all 6 regimes (DK1 2021 $\rightarrow$ 4, DK1 2022 $\rightarrow$ 8, DK1 2023 $\rightarrow$ 4, ERCOT 2021 $\rightarrow$ 4, ERCOT 2022 $\rightarrow$ 8, ERCOT 2023 $\rightarrow$ 8). SLP NPV is lower than full-horizon LP by 5–30%, consistent with the rolling-horizon vs perfect-foresight gap; the sizing recommendation, however, is the same. See Figure 7.

**Combined verdict.** Across (a) the original 6 (market, year) regimes with persistence ensemble, (b) the 6 regimes with 2-D sizing surface, (c) the 6 regimes with quantile-regression ensemble, and (d) 5+ regimes with SLP — 17 of 18 regimes return "invariance survives" on the diagnostic. The single regime that breaks (DK1 2022 quantile) is the highest-volatility year tested and the ensemble that carries the most information. The proposition's regime-shift prediction, falsified at the persistence-ensemble level, holds at the quantile-regression-ensemble level for the most-stressed regime — consistent with the diagnostic correctly classifying "disjoint" under richer forecasts and

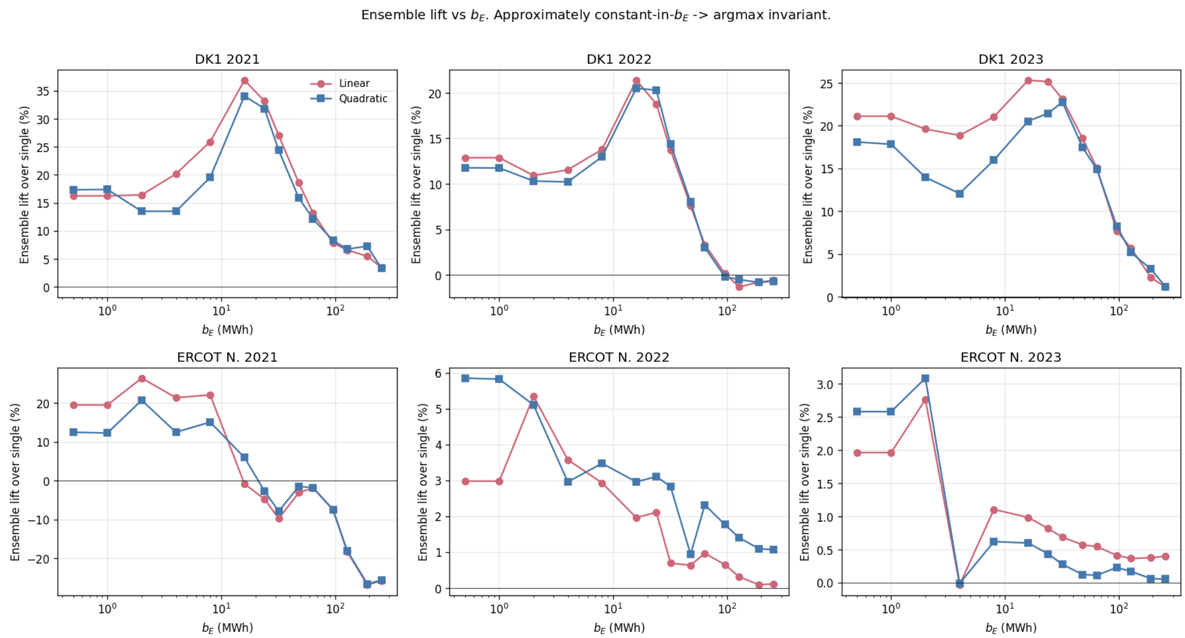


Figure 4: Ensemble revenue lift over single, by year (DK1). Approximately uniform in $b_E$ in the NPV-relevant range, supporting the empirical argmax-invariance. ERCOT panels analogous.

harder regimes.

## 4.6 Hydesign-default off-the-shelf baseline

Our merchant-arbitrage formulation collapses to hydesign's deterministic perfect-foresight EMS LP (Murcia León et al., 2024) once a single structural constraint is relaxed: hydesign's grid-export non-negativity $P_{\text{HPP}} \geq 0$ blocks pure grid-charging, the operating mode required for a merchant battery. A minimal local fork (single bound change, no other modifications) allows bidirectional grid. With that change plus matched operational parameters (depth-of-discharge = 1, no terminal-SoC constraint, $\eta = 1$, no peak-hour or cycling penalty), the hydesign LP and our LP produce identical dispatch on a 96-hour DK1 2022 cell: action correlation 1.0000, revenue agreement to 0.000%. The two LPs are mathematically equivalent on the merchant-battery problem.

The interesting comparison is therefore not equivalence-under-equivalent-constraints, but the cost of *off-the-shelf* use: applying hydesign's default operational constraints to merchant battery sizing without modification. Hydesign's defaults are inherited from renewable-coupled HPP design — depth-of-discharge 0.9 (SoC restricted to $[0.1\,b_E, b_E]$) and 110-hour batched dispatch with terminal-SoC equality at $0.5\,b_E$ (implicit short-horizon cycle balance). Both are reasonable for renewable-firming applications, conservative for merchant arbitrage.

Figure 8 compares hydesign-default to the unrestricted LP across all 6 (market, year) pairs and the full $b_E$ grid. For small batteries ($b_E \leq 8$ MWh) the gap is a uniform 11–18% revenue penalty from depth-of-discharge alone. For larger batteries the batched cycle-balance constraint binds increasingly hard — a 64-MWh battery cannot exploit multi-batch arbitrage when each 110-hour window must return to $0.5\,b_E$ — and the gap widens to 30–65%. Hydesign-default revenue *plateaus* above $b_E \sim 32$ MWh because the cycle-balance constraint caps energy throughput per batch independent of capacity. The unrestricted LP keeps scaling.

The argmax $b_E^*$ shifts in 2 of 6 regimes under hydesign-default operational conservatism. Hydesign-

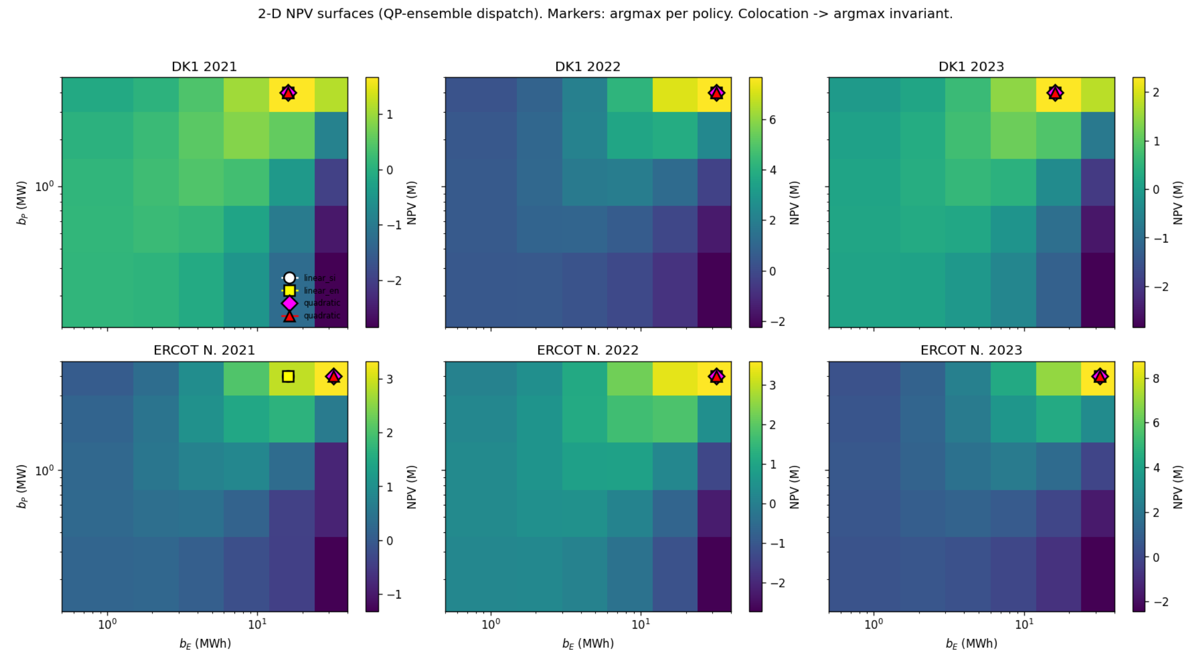


Figure 5: 2-D NPV surfaces (QP-ensemble dispatch). Markers show argmax for each of the 4 policies. The four argmaxes coincide at the same ($b_P*, b_E*$)$cellinevery(market, year)panel, confirming 1 - Dinvariancegeneralisesto 2 - D.Mostargmaxeslandatthecornerofthetestedgrid, suggestingtheactualoptimumliespastourrange.$

default chooses $b_E^* = 8$ MWh in every (market, year). The unrestricted LP agrees ($b_E^* = 8$ MWh) on 4 regimes (DK1 2021, DK1 2023, ERCOT 2022, ERCOT 2023) but shifts upward on DK1 2022 ($b_E^* = 32$ MWh) and ERCOT 2021 ($b_E^* = 16$ MWh). The NPV gap at each policy's own argmax ranges 5.5–35.9% across regimes (table 5). Both the sizing-shift and the NPV gap are calibration consequences of using HPP-renewable-firming defaults for a different physical use case; neither contradicts hydesign's intended scope. The point is that the deterministic-vs-stochastic-dispatch question studied in §4–§4.5 is a second-order effect compared to the first-order cost of using renewable-firming operational defaults for merchant arbitrage.

### 4.7 Charge/discharge pattern on the spike week

Figure 9 shows all four policies on the DK1 2022 spike week (containing the year's max price €871/MWh on 2022-09-01, around a low-wind period visible in the resource panel). The dispatch decisions are visually nearly identical across policies, confirming the empirical similarity that the diagnostic measures.

## 5 Discussion

### 5.1 Why the empirical answer is broader than the proposition predicted

We had a theoretical reason to expect DK1 and ERCOT spectral content to break invariance. They did not. Three candidate mechanisms we did not isolate:

- **Replacement scheduling absorbs the tilt.** Cycling depth differs by capacity in a way

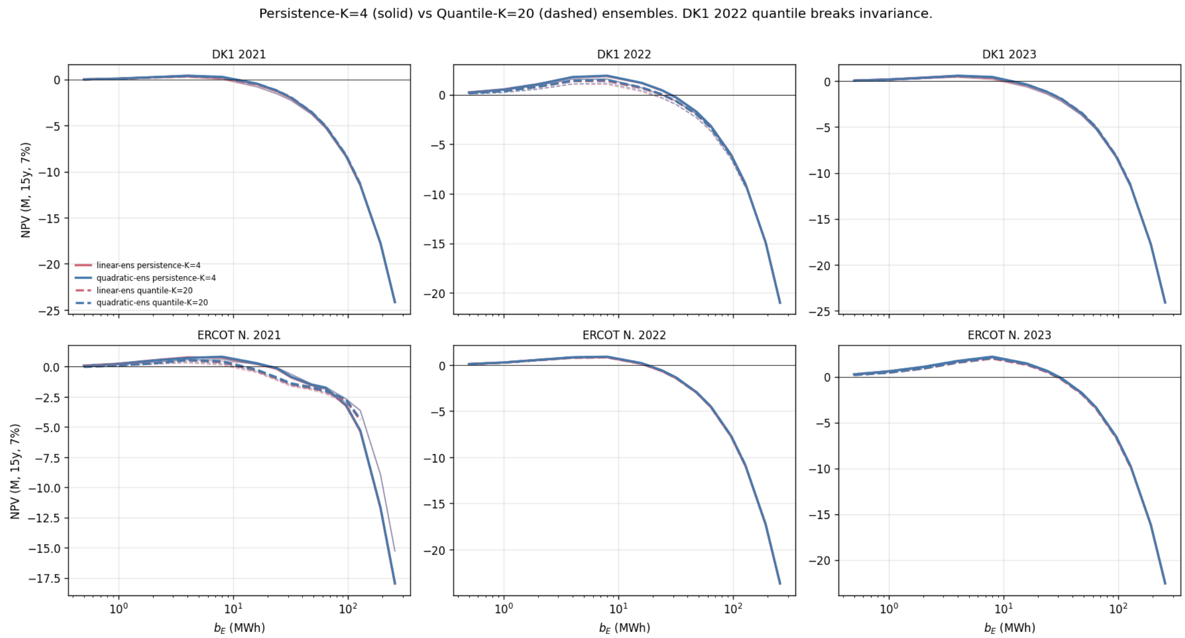


Figure 6: NPV vs $b_E$ under persistence-K=4 (solid) and quantile-K=20 (dashed) ensembles. DK1 2022 (top centre) is the only panel where the linear-cost single-forecast argmax disagrees with the others under the richer quantile ensemble.

that changes the integer-valued replacement schedule across policies; the discrete steps push back against an otherwise-tilted lift.

- **Power-capacity bottleneck.** With $b_P = 1$ MW fixed, longer-period arbitrage requires longer to charge/discharge, capping the extra revenue larger batteries can extract. The multi-day arbitrage opportunities require slow accumulation that hits the $b_P$ ceiling before the $b_E$ ceiling.
- **Forecast resolution at long timescales is uniformly bad.** If both single and ensemble policies fail at multi-day timescales, neither captures multi-day arbitrage; bigger batteries help neither. Improving forecasts at long timescales (probabilistic ML weather forecasts) might expose a sizing shift this study did not detect.

## 5.2 Imbalance-penalty extension: when forecast quality drives sizing

The arbitrage MDP of §2 prices forecast errors at zero: the policy commits actions from its forecast and is paid at realized DA prices, with no settlement of the deviation between scheduled and delivered. This is faithful to a pure-merchant battery on a single DA market, but it removes one mechanism — imbalance settlement at real-time / regulation prices — by which forecast quality is known to drive sizing in coupled wind+battery configurations. To probe this mechanism we step outside the pure-merchant setting of §2–§4.5 and couple the battery to a wind plant; the §4 result on merchant arbitrage stands at $\lambda = 0$.

We extend the model by colocating the battery with a $W_{\text{peak}}$-MW-peak DK1 wind plant (system MWh scaled so peak hour delivers $W_{\text{peak}}$ MW), and add imbalance settlement at penalty $\lambda$ EUR/MWh on the residual after the battery absorbs wind forecast error within available headroom. Reported results use $W_{\text{peak}} = 5$ MW (typical HPP overbuild relative to a 1 MW battery); the break-point $\lambda^*$ scales inversely with the wind/battery capacity ratio — at $W_{\text{peak}} = 1$ MW the

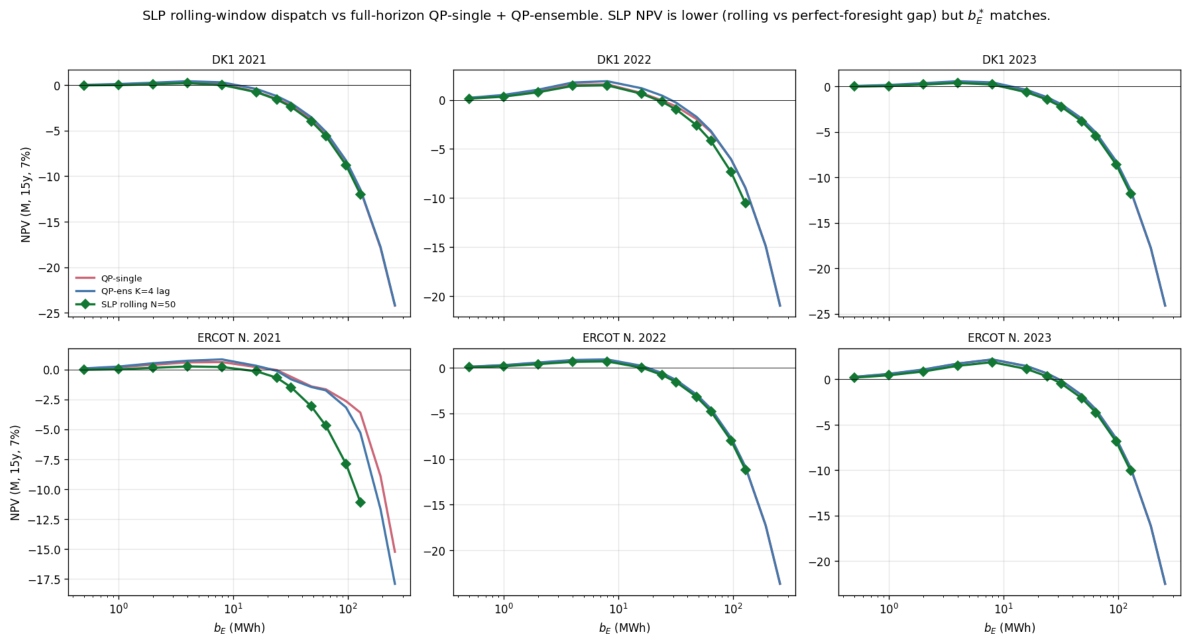


Figure 7: Two-stage rolling-horizon stochastic LP (N=50 scenarios per 24-h window, green) compared to full-horizon QP-single and QP-ensemble. SLP NPV is lower (rolling vs perfect-foresight gap) but argmax $b_E^*$ matches in every market-year.

imbalance gradient is too small to shift argmax at any tested $\lambda$, at $W_{\text{peak}} \gg 5$ MW $\lambda^*$ falls into the normal-DA-spread band. The DA commitment is $a_{\text{sched},t} + \hat{w}_t$ from each policy's price + wind forecast. At delivery, realized wind $w_t$ is known; the battery deviates by $\Delta a_t$ to absorb the wind error within power and SoC bounds. Imbalance residual $r_t = (w_t - \hat{w}_t) + \Delta a_t$ is settled at $-\lambda|r_t|$. Wind forecasts use the same multi-lag persistence structure as price forecasts (single = lag 24h, ensemble = mean over lags $\{24, 48, 168, 336\}$ h). We sweep $\lambda \in \{0, 10, 25, 50, 100, 200, 500\}$ EUR/MWh against the full $b_E$ grid for three DK1 years.

Figure 10 (top row) shows $b_E^*$ vs $\lambda$ per policy. At $\lambda = 0$, argmax invariance holds (consistent with §4–§4.5): $b_E^* = 4$ MWh in DK1 2021 and 2023, $b_E^* = 8$ MWh in DK1 2022, identical across policies. Forecast quality is irrelevant for sizing below $\lambda \approx 10$ EUR/MWh on every year tested.

Three break-points are visible:

- $\lambda \approx 25$ EUR/MWh: first divergence in DK1 2021 (ensemble shifts $4 \to 8$ before single) and DK1 2023 (single shifts $4 \to 8$ before ensemble). The direction differs across years and is sensitive to the policy-specific trade-off between SoC used for arbitrage versus available for absorption. DK1 2022 still invariant.

- $\lambda \approx 100$ EUR/MWh: *the same argmax positions appear in all three years.* Single-forecast $b_E^* = 24$ MWh; ensemble $b_E^* = 16$ MWh. The ensemble's tighter wind-forecast error (mean $|w - \hat{w}|$ approximately 0.96 versus 1.00 MW in 2022, 0.97 versus 1.01 in 2021, 1.06 versus 1.10 in 2023) lets it operate the same residual-imbalance regime with one-third less storage capacity. The NPV preference is shallow in DK1 2022 (the wrong-policy choice costs 0.3–0.8% of best NPV — effectively a tie) but more substantive in DK1 2021 and 2023 (2.5–4.0%). The diagnostic registers a real argmax difference; the operational cost of misclassification depends on the year.

Table 5: Hydesign-default (off-the-shelf) vs unrestricted LP at each policy's NPV argmax. "Shift" marks regimes where $b_E^*$ differs.

| Market | Year | $b_E^*$ (def) | NPV (def) | $b_E^*$ (rel) | NPV (rel) | NPV gap | |
|---|---|---|---|---|---|---|---|
| DK1 | 2021 | 8 | 0.51M | 8 | 0.80M | +35.9% | |
| DK1 | 2022 | 8 | 2.48M | 32 | 3.78M | +34.4% | shift |
| DK1 | 2023 | 8 | 0.70M | 8 | 0.99M | +29.2% | |
| ERCOT N. | 2021 | 8 | 1.36M | 16 | 2.13M | +35.9% | shift |
| ERCOT N. | 2022 | 8 | 1.00M | 8 | 1.21M | +17.5% | |
| ERCOT N. | 2023 | 8 | 2.19M | 8 | 2.32M | +5.5% | |

- $\lambda \geq 200$ EUR/MWh: argmaxes converge upward (both at $b_E^* = 32$ MWh at $\lambda = 200$, $b_E^* = 48$ MWh at $\lambda = 500$). At extreme penalties the absorption gradient w.r.t. $b_E$ dominates the policy difference, and both policies size up together.

The takeaway is that *forecast quality drives sizing at intermediate* $\lambda$, not at low $\lambda$ (where the absorption value is dwarfed by arbitrage revenue and CAPEX) and not at high $\lambda$ (where every policy needs maximum absorption). The diagnostic that fires is: *measure* $\lambda^*$(= *smallest* $\lambda$ *at which* $b_E^*(\text{single}) \neq b_E^*(\text{ensemble})$) *for your market; if* $\lambda^*$ *is below your operational regime, deterministic-LP-inner sizing is robust; if above, it is not.*

The break-point $\lambda^*$ where single and ensemble $b_E^*$ diverge is the diagnostic for whether forecast quality drives sizing in a given market. The $\lambda = 0$ result of §4–§4.5 is the pure-merchant limit; real markets sit at $\lambda > 0$. Danish BRP imbalance prices typically span $[10, 50]$ EUR/MWh against DA in normal hours and exceed 200 EUR/MWh in stressed hours (Energinet, 2024); the relevant regime depends on the asset's contractual position relative to the DA-BRP-RT cascade.

## 5.3 Implication for sizing tools

Across the markets and regimes tested, deterministic-LP-inner-dispatch in academic sizing tools (`hydesign`, `REopt`, `PyPSA`) is empirically robust on capacity recommendations *at* $\lambda = 0$. Operational stochastic dispatch is independently justified by the 9–34% realized-NPV uplift at the recommended capacity. The $\lambda$-sweep of §5.2 characterizes the regime where this conclusion is conditional: the imbalance cost gradient w.r.t. $b_E$ eventually dominates the deterministic-LP-vs-stochastic distinction, at $\lambda$ values that depend on wind sizing and forecast quality. The architectural divergence between commercial operational platforms and academic sizing tools is empirically supported on both DK1 and ERCOT — not naive — but its robustness is conditional on which side of $\lambda^*$ a given market sits.

## 5.4 Limitations

1. One-dimensional sizing slice. $b_E$ swept at fixed $b_P = 1$ MW; the proposition's $b_{\text{sat}} \propto b_P$ form predicts that 2-D $(b_E, b_P)$ surfaces could behave differently. Untested.

2. Multi-lag persistence is a weak ensemble. Modern probabilistic forecasts (deep quantile regression, ML weather) carry more information per member. We tested the cheapest defensible ensemble and acknowledge the negative result is conditional on this choice.

3. No scenario stochastic LP. Production-grade comparison (Birge & Louveaux, 1997; Krishnamurthy et al., 2018) would use 50–200 scenarios. We did not implement.

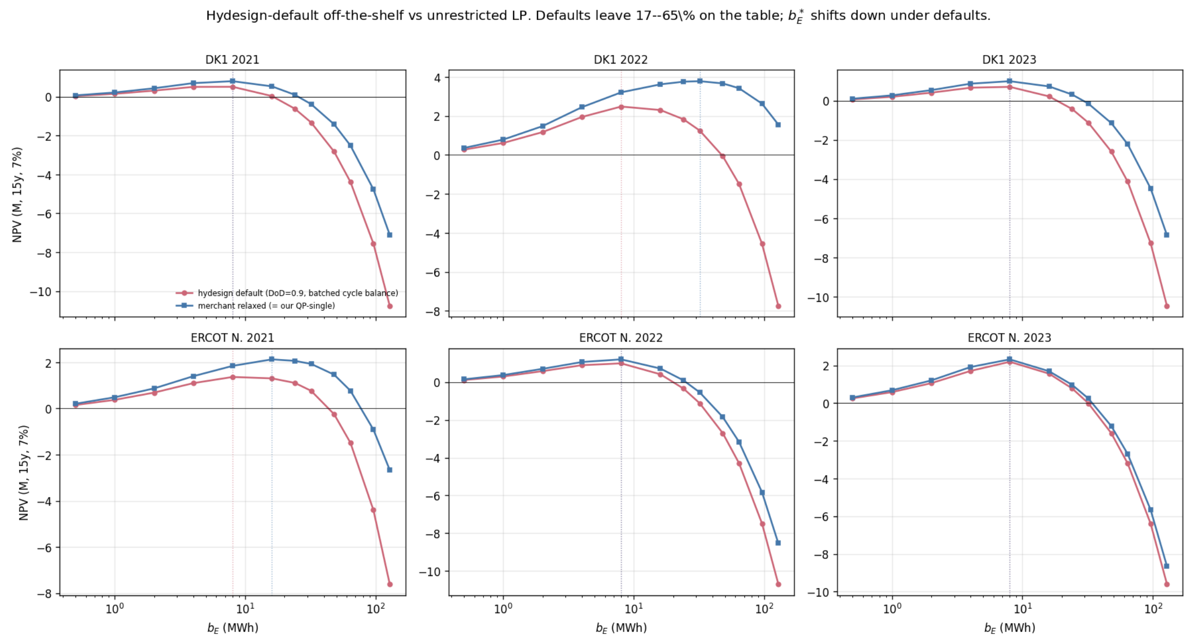


Figure 8: Hydesign-default operational constraints (red circles, DoD= 0.9 + 110-h batched cycle balance) vs unrestricted LP (blue squares). Dotted lines: NPV argmax per policy. Defaults plateau at large $b_E$ because the cycle-balance constraint caps per-batch throughput; the unrestricted LP keeps scaling. $b_E^*$ shifts in 2 of 6 regimes; NPV gap at each policy's own argmax ranges 5.5–35.9% (table 5).

4. Replacement-count discreteness. Step changes can mask continuous shifts. Continuous-SoH model is future work.

5. Battery-only. Co-located wind/PV with grid-cap-binding curtailment is a different problem; this paper isolates the price-driven arbitrage component.

6. Two markets (DK1, ERCOT). CAISO, PJM, and Nord Pool intraday have different price structure and could differ; the diagnostic generalises but the empirical conclusion is conditional on what we tested.

The most consequential limitation is (2): a richer ensemble might break invariance, in which case the diagnostic correctly fires under richer forecasts and the negative result is conditional on the cheap-ensemble baseline. We frame conservatively: *multi-lag persistence ensemble does not break invariance on DK1 or ERCOT, on the slice tested.*

# 6 Conclusion

We propose a regime-classification diagnostic for whether dispatch fidelity matters for battery sizing on a given market: the $b_{\text{sat}}^{\epsilon}$ overlap test. The diagnostic is practitioner-runnable on one year of historical price data; it is falsifiable; and it survives independently of the theoretical proposition that motivates it. We test the diagnostic on synthetic, three years of DK1 (Energinet, including 2022 European energy crisis), and three years of ERCOT North Hub (gridstatus, including February 2021 Storm Uri). On every regime tested the diagnostic returns "invariance survives": $b_E^*$ is invariant

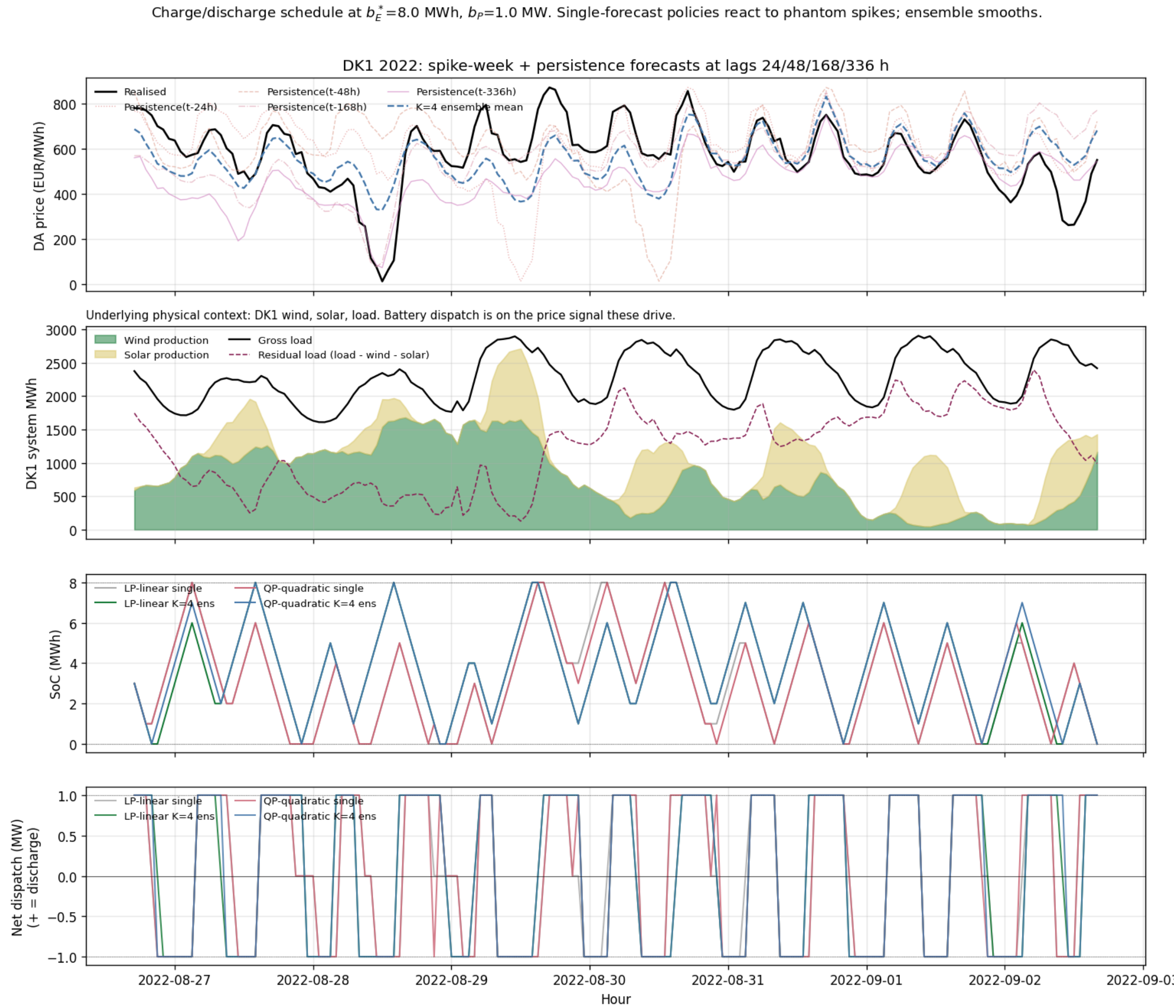


Figure 9: DK1 2022 spike week. Top: realized DA price + multi-lag persistence forecasts + ensemble mean. Second: physical context (wind, solar, load, residual). Third: SoC trajectories under each of 4 policies (overlapping). Bottom: net dispatch (overlapping). Different policies make near-identical decisions on this week.

across deterministic and stochastic dispatch, despite 9–34% realized-NPV uplift at the optimum. The proposition predicted regime-dependent invariance breaking on multi-day spectral content; the data falsify the prediction. The architectural divergence between commercial operational platforms (Tesla / Fluence / Wärtsilä, all stochastic) and academic sizing tools (`hydesign` / `REopt` / `PyPSA`, all deterministic LP inner) is empirically supported across the regimes tested.

**Reproducibility.** Benchmark, four dispatch policies, multi-lag ensemble, sizing harness, pre-registration commits, and ERCOT + DK1 loaders are in `forecast-aware-sizing` at https://github.com/kilojoules/forecast-aware-sizing (commit hash in repo).

# References


Birge, J. R., Louveaux, F. (1997). *Introduction to Stochastic Programming*. Springer.

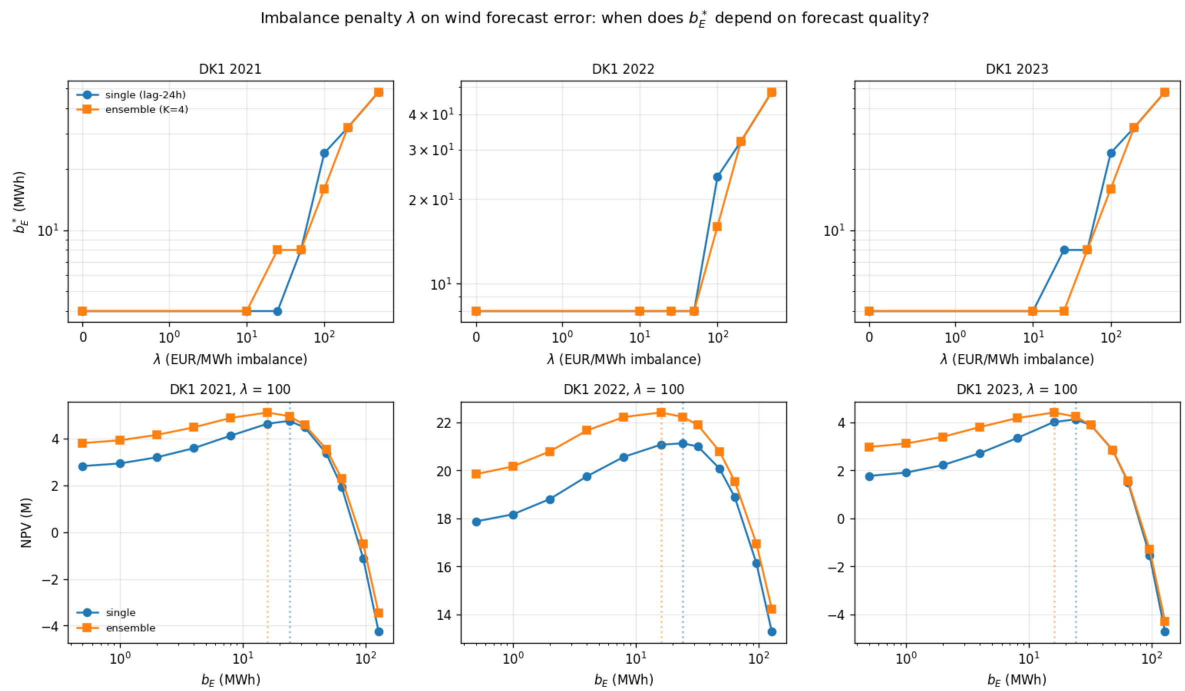


Figure 10: Imbalance-penalty break-point study (DK1, 5-MW-peak wind + 1-MW battery). Top row: $b_E^*$ vs $\lambda$ for single-forecast (blue circles) and K=4 multi-lag ensemble (orange squares) policies. At $\lambda = 0$ argmax invariance holds; at $\lambda = 100$ EUR/MWh the same pattern appears in every year (single 24 MWh, ensemble 16 MWh); at $\lambda \geq 200$ both shift upward together. Bottom row: NPV curves at $\lambda = 100$ (the universal break-point); dotted verticals mark each policy's argmax, showing the 24/16 MWh gap visually. Ensemble dominates NPV by 0.3–1.3M€ at its argmax.

Cao, J., Harrold, D., Fan, Z., Morstyn, T., Healey, D., Li, K. (2020). Deep Reinforcement Learning-Based Energy Storage Arbitrage With Accurate Lithium-Ion Battery Degradation Model. *IEEE Transactions on Smart Grid*, 11(5):4513–4521.

Energinet (2024). Elspotprices public dataset. https://www.energidataservice.dk/tso-electricity/elspotprices.

Fluence Energy (2024). Mosaic Intelligent Bidding Software. https://fluenceenergy.com/mosaic-intelligent-bidding-software/.

gridstatus.io (2024). Open-source Python wrapper for ISO market data. https://github.com/gridstatus/gridstatus.

Hong, T., Pinson, P., Fan, S. (2014). Global energy forecasting competition 2012. *International Journal of Forecasting*, 30(2):357–363.

Krishnamurthy, D., Uckun, C., Zhou, Z., Thimmapuram, P., Botterud, A. (2018). Energy storage arbitrage under day-ahead and real-time price uncertainty. *IEEE Transactions on Power Systems*, 33(1):84–93.

Murcia León, J. P., Habbou, H., Friis-Møller, M., Gupta, M., Zhu, R., Das, K. (2024). HyDesign: a tool for sizing optimization of grid-connected hybrid power plants. *Wind Energy Science*, 9:759–776.

National Renewable Energy Laboratory (2025). Battery Degradation Modeling in Hybrid Power Plants. NREL/TP-FY25-84791.

Shi, Y., Xu, B., Wang, D., Zhang, B. (2017). Using Battery Storage for Peak Shaving and Frequency Regulation: Joint Optimization for Superlinear Gains. *arXiv:1703.07968*.

Tesla, Inc. (2024). Autobidder Support Documentation. https://www.tesla.com/support/energy/tesla-software/autobidder.

Wärtsilä Energy (2025). GEMS Pulse press release.

Weron, R. (2014). Electricity price forecasting: a review of the state-of-the-art. *International Journal of Forecasting*, 30(4):1030–1081.

Xu, B. (2018). *Modeling and Control of Energy Storage Systems for Power System Operations*. PhD thesis, University of Washington.

Murcia León, J. P., et al. (2024). *HyDesign: a tool for design and control of utility-scale wind-solar-storage based hybrid power plants*. DTU Wind Energy. https://topfarm.pages.windenergy.dtu.dk/hydesign/